

Assessing climate risks from satellite imagery with machine learning: A case study of flood risks in Jakarta

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ABSTRACT

Consistent and timely assessment of climate risks is crucial for planning disaster mitigation and adaptation to climate change at the local community level. This article presents an automatized method for monitoring climate risks with machine learning on satellite imagery, specially targeting riverine and coastal floods. Our research demonstrates that disaster-related risk measurement becomes more comprehensive and multi-faceted by including the following components: hazards, exposure, and vulnerability. Our model first maps hazard-related risks with geo-spatial data, then extends the model to incorporate exposure and vulnerability. In doing so, we adopt a clustering-based supervised algorithm to sort satellite images to produce the climate risk scores at a grid-level. The developed model was tested over multiple ground-truth datasets on flood risks in the region of Jakarta, Indonesia. Results confirm that our model can assess climate risks in a granular scale and further capture potential risks in the marginalized areas (e.g., slums) that were previously hard to predict. We discuss how computational methods like ours can support humanitarian projects for developing countries.

1. Introduction

The climate crisis is a pressing global challenge. The IPCC report (2022) urged immediate action to limit the global temperature rise to 1.5 °C; beyond which point, diverse climate drivers and hazards will interact to produce unprecedented magnitude and frequency of extreme compound events (Scott, 2016; Zscheischler et al., 2018). Consequently, climate risk assessment is becoming crucial for planning disaster mitigation and adaptation (Jones, 2001; UNDRR, 2019). However, existing risk and disaster management relies on resource-intensive surveys that are subject to increasing concerns over insufficient, incomplete, and inconsistent local-level assessment (Adger et al., 2018). Risk assessment has also been confined to national- and international-level climate change policies (e.g., López Royo et al., 2016) and has not adequately presented over multiple-temporal scales (Arribas et al., 2022).

With the rapid development of artificial intelligence (AI) and the increasing availability of digital data, machine learning models are expected to bring breakthroughs in risk assessment (Rolnick et al., 2022; Zennaro et al., 2021). For instance, computer vision techniques or AI methods that train machines to interpret visual images are being used to infer the level of climate risks in distant parts

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of the world from satellite images. The combination of machine learning and remote sensing augments labor-intensive and time-consuming surveys with faster and more accurate digital assessment. Satellite imagery provides a bird's-eye view of wide geographical areas, making it a viable tool for monitoring hazards. Also, geographic information system (GIS) datasets provide a consistent and standardized view of the Earth's surface, allowing for continued and repeated analysis that can detect temporal changes. Numerous advantages of remote sensing, such as extensive coverage, high resolution, timeliness, cost-effectiveness, and consistency, provide innovative and indispensable tools for climate risk assessment, thereby contributing to global humanitarian efforts to tackle climate change.

Some studies determine the damage extent of natural disasters depicted in satellite images to measure climate risks with machine learning (Kim et al., 2022; Mateo-García et al., 2021). Such techniques cover diverse types of disasters, including floods (Antzoulatos et al., 2022), droughts (Hakam et al., 2023), and landslides (Mohan et al., 2021). Other studies have applied change detection algorithms for assessing infrastructural damage (Wu et al., 2021; Xie et al., 2020). Herein, algorithms identify substantial differences in the condition of buildings, roads, and other artificial infrastructures impacted by natural disasters. A seminal work by Gupta et al. (2019) constructed the xBD dataset to detect building footprints in pre- and post-events using satellite images and to assess the degree to which buildings were destroyed.

In addition to these efforts, our model produces multi-faceted climate risk scores, not only assessing damage from specific events but also the potential likelihood of receiving damages. Our model considers social dimensions such as susceptibility, sensitivity, and adaptability to climate change, besides the physical dimensions of natural disasters (Mokrech et al., 2012). This approach benefits from the state-of-the-art computer vision techniques that infer socio-economic indicators from satellite imagery (e.g., poverty and income; Ahn et al., 2023; Jean et al., 2016; Yeh et al., 2020). Likewise, AI-driven methods can be applied to heterogeneous remote sensing datasets across global regions to spot highly-vulnerable areas to climate change.

Our climate risk assessment model measures three commonly discussed components of climate risks: hazards, exposure, and vulnerability. We adopt their definitions given by the IPCC report (2022), which defines hazards as “the potential occurrence of a natural or human-induced physical event”, exposure as “the presence of people and other assets,” and vulnerability as “the propensity or predisposition to be adversely affected.” While hazards were adequately assessed by the previous approaches, this article adds the other two components to the assessment model by predicting ‘the presence of asset’ for exposure (e.g., population density) and ‘the level of wealth’ for vulnerability in the affected areas (e.g., the level of expenditure).

We illustrate the model with a case study on Jakarta, the capital of Indonesia. This region serves as a good case study because of its extremely high risks. Jakarta is one of the largest cities in the world with a population of over eleven million. It was ranked as the world's most vulnerable city to environmental hazards in terms of climate change, pollution, and natural disasters (Verisk Maplecroft, 2021). The city is located on a flat alluvial area with an elevation less than 20 m above sea-level and is surrounded by thirteen rivers (Firman et al., 2011; Simarmata and Surtiari, 2019). The estimated global sea-level rise is about 70 to 100 cm by 2100 under RCP 8.5 (IPCC, 2022) and up to 95 percent of its coastal areas and one-third of the capital could be submerged by 2050. The risks are exacerbated by human-made factors such as dense land-use development, an overburdened drainage system, excessive groundwater extraction, and land subsidence (Abidin et al., 2001; 2011; Chaussard et al., 2013). We utilize the high-resolution daytime satellite imagery at a resolution of 1.193 m per pixel to detect exposure and vulnerability in terms of assets and wealth.

Climate risks in Jakarta had been evaluated within two distinct contexts with the most severe threats (IPCC, 2022:1464): *riverine floods*, influenced by rainfall trends, and *coastal floods*, attributed by sea-level rise, high tides, storm surges, and man-made land subsidence. Both types of floods present unique effects and mechanisms, compounded by climate change (Ghanbari et al., 2021). To assess the combined impact of these flood types, *hazard scores* for both riverine and coastal floods are integrated for Jakarta. Hazards are assessed using existing geo-spatial data on physical events, recognizing that not all events are captured by satellite imagery. The level of riverine floods is determined through a flood modeling framework, while coastal floods are analyzed using a digital elevation

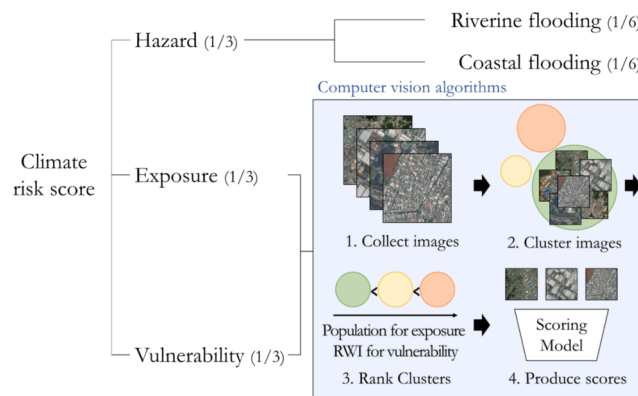


Fig. 1. The proposed model for assessing three components of climate risks. Hazards are calculated from GIS (geographic information system) data on floods. The other components are measured by computer vision algorithms with four steps: collecting images, clustering images, ranking clusters based on population and relative wealth index as training datasets, and producing scores. Satellite images were downloaded from the ESRI ArcGIS REST API.

model.

The calculated climate risk scores offer information for decision-makers to develop effective climate change adaptation and mitigation strategies. The scores enhance the risk assessment by considering other factors besides the frequency and intensity of natural disasters. Namely, the scores identify the socio-economic conditions that contribute to climate-related risks and opportunities. The case study of Jakarta demonstrates that the marginalized citizens, such as slum dwellers, are at the greatest risk of the impacts of riverine and coastal floods. Our scores indicate that informal settlements have a moderate probability of experiencing hazards but have higher exposure and vulnerability scores compared to other urban areas. This conclusion may not be reached with existing approaches that only focus on hazard aspects of natural disasters.

The remainder of this article is organized as follows. Section 2 introduces the dataset used for the analysis. Section 3 presents the methods to measure hazards, exposure, and vulnerability from satellite imagery and remote sensing methods. In Section 4, we present the results based on the case study of Jakarta and evaluate the model against other baselines. The section also shows the regression analysis to understand how climate risks are unevenly distributed across the city. Section 5 concludes.

2. Dataset: Daytime satellite imagery

We present a methodology to evaluate *hazards*, *exposure*, and *vulnerability* at a granular level (Fig. 1). The spatial units are zoom-level 17 grids with a pixel resolution of 1.193 m. The zoom level resolution system refers to the number of 256-by-256 pixel tiles (or grids) required to display a specific area on a map. In this system, the world is divided into tiles, with each tile representing a distinct geographic region. At zoom level 0, the entire earth is displayed on a single tile, resulting in a coarse image. As the zoom level increases, the number of tiles required to display the map increases exponentially, and the displayed image becomes more detailed. For example, at zoom level 1, the world is divided into 2x2 tiles, with each tile representing one-fourth of the earth's surface. At zoom level 17, each of the entire earth's surface is subdivided into $2^{17} \times 2^{17}$ tiles, with each tile now representing an area of 0.09 km^2 . This level of zoom resolution enables agile community-driven assessment and policymaking (see Table 1 for examples of zoom level 17 tiles). Satellite images can be accessed through various publicly available or privately-owned platforms. The World Imagery dataset, accessible via the ESRI ArcGIS REST API (https:// tiledbasemaps.arcgis.com/arcgis/rest/services/World_Imagery/MapServer), is one such data source.

We collected satellite images of a wider area covering Jakarta (i.e., metropolitan area), spanning the total number of tiles of 76,710 taken from 2020 for the three visible bands (i.e., red, green, and blue). However, the results were shown within Jakarta only; Jakarta comprises six counties: Central, East, North, South, West Jakarta, and the Thousand Islands. The small island areas, although more susceptible to coastal floods, were excluded from the analysis to focus on the risks in the inland. As the output of the algorithm, each grid in Jakarta is assigned three scores: hazards, exposure, and vulnerability. The exposure and vulnerability scores are computed using satellite imagery, with the size of each grid determined by the zoom resolution. For the hazard score, geo-spatial datasets are rescaled to zoom level 17 for compatibility with the other scores. Each component is assigned a score between 0 and 1, with an equal weight that comprises the climate risk score, representing the average of the three component scores. There can be numerous ways to assign weights to the three risk components (Peng and Zhang, 2022), yet this work employs the basic average and considers other alternatives as future directions. Once the grid-level climate risk scores were computed, we aggregated them into 262 districts in Jakarta by averaging the grid scores.

3. Methods

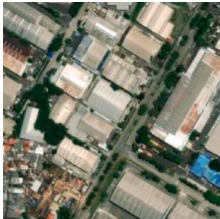
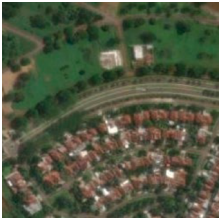
3.1. Climate risk assessment model

Our model (depicted in Fig. 1) consists of two parts: the first is for hazard mapping, and the second is for measuring exposure and vulnerability. While the IPCC report defines hazards as a combination of natural and human-induced physical events, we here limit to two natural events considered crucial by numerous empirical studies (e.g., Firman et al., 2011) in Jakarta: geo-spatial data related to riverine and coastal floods.

We employed the existing riverine flood hazard scores derived from the SOBEK model. SOBEK, a comprehensive modeling framework developed by Deltares (2024), is designed for flood forecasting. This framework integrates variables related to river discharge, canal conditions, and sewer systems, as well as other hydrological and hydraulic indicators. It simulates complex flows and water-related processes in both one-dimensional and two-dimensional systems for major rivers. The model utilizes grid cells with dimensions of $50 \times 50 \text{ m}$ for the Ciliwung floodplain and $100 \times 100 \text{ m}$ for other areas of Jakarta (Budiyo et al., 2015). It is capable of depicting inundation depth and extent for various return periods, including 1, 2, 5, 10, 25, 50, and 100 years. For the purpose of the long-term climate risk assessment, a flood with a 50-year flood return period was chosen to generate the riverine flood hazard map of Jakarta. The pre-calculated dataset was obtained from Budiyo et al. (2015:395).

For the coastal floods, we use the estimate values based on the elevation of grids in ASTER DEM v2 (Abrams et al., 2010), which is based on a 30 m resolution global digital elevation model (GDEM). Although Indonesia has a high-resolution elevation model, DEMNAS (BIG, 2024), with a resolution of less than 10 m, we use the GDEM for developing a universally adaptable model applicable across various countries. We use a one-meter threshold above sea-level to calculate its hazards. This threshold is based on previous studies (Takagi et al., 2016), which indicate that a flood depth of 1 m or more poses significant threats and substantial damage to human populations. Reflecting this threshold, the hazard score was given as zero for places with a height above a meter. Conversely, areas with current elevations of 0 m or below were given a score of 0 to reflect the highest risk. Other areas between 0 and 1 m was

Table 1
Selected example clusters and their socio-economic conditions. Out of 78,037 satellite image tiles covering Jakarta, a deep learning algorithm identified 20 clusters based on the visual similarity of images. We show a single example image for five of the clusters, where cluster numbers are assigned randomly. Each grid tile can be assigned exposure and vulnerability scores, based on existing statistics such as the population density from the WorldPop dataset and the relative wealth index from the RWI dataset. Satellite data shown have been downloaded from the ESRI ArcGIS REST API (https://tiledbasemaps.arcgis.com/arcgis/rest/services/World_Imagery/MapServer).

Cluster Number	5	2	14	17	10
ExampleGrid Image					
Features(Human Annotation)	Highly developed urban area	Highly dense residential area	Modernized rural villages	Traditional rural village	Uninhabited
Population Density (WorldPop)	13,609 / km ²	14,233	5,130	6,295	735
Relative Wealth Index (RWI)	1.24	1.22	0.84	0.79	−0.48

converted to a hazard score between 0 and 1, accordingly. While previous research highlights the importance of including land subsidence effects to understand future coastal flood risks (Abidin et al., 2011; Takagi et al., 2016), this study excludes such effects to focus on calculating the current level of hazards and climate risks in Jakarta.

To compute the exposure and vulnerability scores, we adopt a cluster-based supervised approach for analyzing satellite images (Han et al., 2020b). This work groups satellite images into meaningful *clusters* based on the similar visual features detected by the Deep Cluster algorithm (Caron et al., 2018). Satellite images of covering Jakarta metropolitan area were grouped into 20 clusters based on the same algorithm, which is two orders of magnitude scaled-down version from the total of 76,710 images. Instead of involving human labeling of the scaled down clusters as in Han et al. (2020b), we use existing grid-level proxies of exposure and vulnerability to further automate the process. Given that exposure quantifies people and assets, one may consider utilizing population density or the presence of human artifacts. Past studies have confirmed a significant positive correlation between population density and other asset related indexes such as building footprint, market size, and the number of establishments (Ahn et al., 2023; Han et al., 2020b). If a region has a high population density, it indicates a greater concentration of people and assets in that area; consequently, such areas are more susceptible to climate risks in terms of having higher exposure. For vulnerability, a wealth index can represent the propensity to be adversely affected by hazards (Leichenko and Silva, 2014). If a region has a high level of wealth, it will be less vulnerable to the adverse effects of hazards due to climate risks.

We use the population density data with a 100-meter resolution from WorldPop (2024) as a proxy of exposure and the Relative Wealth Index (RWI) with a zoom level 14 resolution (a pixel resolution of 9.555 m) published by Columbia University and Meta (2024) as a proxy of vulnerability. We extrapolated population and wealth indexes to match the zoom level of 17. Given that tiles contain multiple instances of these measures, we take the average population and wealth index for each tile, as shown in Fig. 2. The rationale for adopting WorldPop and RWI as proxies is that they offer grid-level data with the highest resolution available globally among existing datasets. Considering the majority of statistics provided at the district-level, it is difficult to match grid-level satellite imagery with such district-level data (Han et al., 2020a). Moreover, WorldPop and RWI include data for 234 and 135 countries respectively. This research method can be easily applied to other regions without using heavyweight labels and human annotation efforts.

Once population density and RWI scores are assigned, we compute their average scores at the cluster level to learn the machine to produce exposure and vulnerability scores for each grid tile. Table 1 depicts example tiles from five of the clusters, which range from urban areas (clusters 5 and 2) to more rural areas (clusters 14, 17, and 10). Population density and RWI-based ordering may not align; for example, cluster 2 displays a higher population density due to its residential areas, yet cluster 5 is considered to have marginally higher wealth due to its industry presence. In the rural examples, cluster 17 has a higher coverage of building footprints (hence a larger population), yet cluster 14 has a higher RWI score due to its modernized village and regular structures. Next, for all grid images in each cluster, we sort all clusters by the two measures as follows:

$$(higherWorldPop) 2 > 1 > 5 > 6 > 0 > 3 > 4 > 11 > 12 > 17 > 14 > 15 > 16 > 9 > 8 > 13 > 18 > 7 > 10 > 19(lower) \quad (1)$$

$$(higherRWI) 5 > 3 > 0 > 6 > 2 > 4 > 1 > 11 > 12 > 14 > 17 > 16 > 15 > 13 > 8 > 9 > 18 > 7 > 10 > 19(lower) \quad (2)$$

Instead of directly using the population density and RWI scores for each tile, we use them as proxy values and recompute the exposure and vulnerability scores as a continuous scale between 0 and 1. We use rank of each cluster and use a ranker function from Han et al. (2020b) as follows. Let two partially ranked clusters be R_i , where it is defined as $\{C_1, C_2, \dots, C_{20}\}$ indicating C_j is a j -th cluster in the ranking. The rank of clusters r_j should be $r_j = [1, 2, \dots, 20]$. Then, an equal batch size of images is sampled from each cluster C_j . The model produces the concatenated scores for each selected image as S_k in Eq. (3), where $x_{C_j}^k$ refers to the k -th image in cluster C_j .

$$S_k = \text{concat}([f(x_{C_1}^k), f(x_{C_2}^k), \dots, f(x_{C_{20}}^k)]) \quad (3)$$

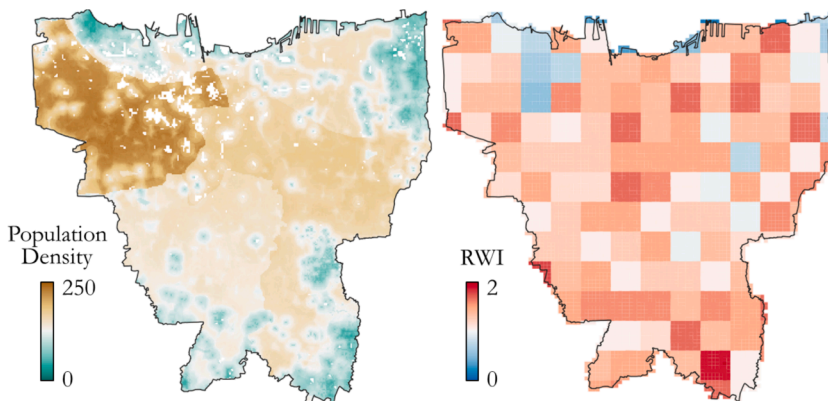


Fig. 2. The training datasets: WorldPop (left) and Relative Wealth Index (right).

However, due to the non-differentiable nature of ranking scores, conventional deep learning models that rely on gradient descent cannot be utilized to maximize this correlation. Therefore, we added the differentiable ranking function to learn approximations of such non-differentiable objective metrics (Engilberge et al., 2019). Finally, the scoring model f trains to match the two rank vectors: $\text{rank}(S_k)$ and r_j . To maximize the performance of the scoring model, the method maximizes the spearman correlation between two vectors with the following equation:

$$\max(1 - \frac{6\|\text{rank}(S_k) - r_j\|_2^2}{m(m^2 - 1)}) \quad (4)$$

3.2. Evaluation and comparison models

To evaluate the performance of cluster-based scoring model, we compare the model with three baseline models. We first consider the Ground-Truth (GT) regression models that directly utilize grid-level WorldPop and RWI scores instead of using any machine learning model or clustering. The model coefficients are calculated by regressing grid-level ground-truth on validation datasets. The second is the Night-Light (NL) regression model whose coefficients are estimated by regressing grid-level nightlight intensity on validation datasets. The nightlight intensity is known to have a correlation with economic activities because it provides a proxy measure of human activity and the presence of artificial lighting (Henderson et al., 2012). This article utilizes the yearly aggregated VIIRS (the Visible Infrared Imaging Radiometer Suite) nightlight data version 2.1 in 2018 from EOG (Earth Observation Group), Payne Institute for Public Policy (Elvidge et al., 2021). Next, we employed pairwise loss as a baseline, which is a commonly used machine-learning approach for learning rankings (Cao et al., 2007; Ahn et al., 2023). The network was trained by optimizing the pairwise loss function, where positive and negative pairs were generated using ground-truth at the grid-level population and RWI.

The evaluation of Jakarta incorporates the following datasets for validation purposes. Although these datasets cover different periods, they can be combined to assess the overall flood risks since the 2010 s. Regarding exposure scores, statistical data and proxies are employed to assess the extent of assets depicted in satellite images. The first dataset utilized is the district-level population density obtained from the Jakarta government's statistics in 2020, as provided by [Open Data Jakarta \(2024\)](#). Additionally, the validation uses the number of evacuees during the January 2013 flood, collected by the Jakarta government (DKI Jakarta, 2013), to assess the actual economic loss resulting from the event. Among the floods for which economic loss data was collected, the 2013 flood was the most severe (Wijayanti et al., 2017) and has the most well-documented data across multiple districts. Therefore, we chose to use the 2013 event for our analysis. Since the data refers to the number of people exposed to physical events, it serves as a verification metric for the level of exposure. These statistics were collected through in-situ survey efforts by the government for 121 districts affected by the flood event. For example, *Rawa Buaya* district, located next to the Cengkareng Drain, experienced the most devastating flood impacts in January of 2013, with a total of 26,628 residents evacuating to shelters due to floods. To normalize the data, the actual numbers were divided by the area of each district.

The third dataset concerns the total area of buildings and the total lengths of roads constructed within the images. To validate these aspects, a manually constructed dataset in 2021 comprising building and road footprints from [OpenStreetMap \(2024\)](#) is employed. Specifically, the floor area of 1,876,830 buildings and the length of 884,084 roads in Jakarta are extracted, and subsequently aggregated at the grid- and district-levels. These aggregated values serve as proxies for assessing economic development, as described by Ahn et al. (2023).

Furthermore, vulnerability scores are validated using municipal statistics related to the level of wealth. These statistics encompass the number of employments in 2020, as provided by [Open Data Jakarta \(2024\)](#), and the amount of government expenditure in 2021, sourced from various publications such as [BPS Jakarta Barat \(2021:38\)](#), [BPS Jakarta Pusat \(2021:102\)](#), [BPS Jakarta Selatan \(2021:34\)](#), [BPS Jakarta Timur \(2022:34\)](#), and [BPS Jakarta Utara \(2021:45\)](#). These two statistics were selected due to their availability at the sub-district level in Jakarta, even though they may not encompass the entirety of the economic landscape. Nonetheless, they serve as valuable proxies for assessing economic prosperity. As for hazard scores, no evaluation is performed since they are based on GIS scenarios that do not need any validation.

3.3. Spatial regression model

When the performance of our model is validated, we utilize the district-level scores as a dependent variable for a regression model to understand how climate risks are unevenly distributed across the city. To remove the spatial autocorrelation and understand the correct spatial distribution of the model, the spatial regression model was applied (Anselin, 2005). Spatial autocorrelation pertains to the relationship between geographically adjacent observations in neighboring locations. These observations are aggregated into polygons representing districts, which may lead to spatial autocorrelation if the spatial units and phenomena do not match, causing an incomplete reflection of the correct spatial distribution. Moran's I is used to quantify spatial autocorrelation, with a positive or negative value indicating similar or contrasting values, respectively, among adjacent observations. The dependent variable's Moran's I value was found to be 0.7, indicating a high level of positive spatial autocorrelation.

We use the contiguity-based method (i.e., queen type) rather than the distance-based method to consider the spatial dependencies from neighboring districts (Anselin, 2005). The independent variables include socio-economic conditions (slum ratio in 2013 and unemployment ratio in 2018; [Open Data Jakarta, 2024](#)) and the proportion of land-use type in 2022 (commercial, industrial, public, religion, medical, education, and green areas; Jakarta Satu, 2024). These variable were chosen to understand which urban areas in

Jakarta are mostly affected by climate risk. Also, the regression included variables of flood facilities: numbers of floodgate, flood post, and pump stations (Pantau Banjir Jakarta, 2024), and lengths of drainage (Jakarta Satu, 2024). Including the variables related to flood facilities would help understand to what extent those mitigation strategies are effective in reducing climate risks in Jakarta. After collecting variables, multicollinearity is tested by comparing variance inflation factors (VIF). Generally, a VIF value above 5 may be of concern (Menard, 2001). Lastly, the dummy variables of counties (Central, East, North, South, and West Jakarta) are included for fixed effects.

The final stage involved selecting a spatial regression model by choosing between a spatial lag model (SLM) and a spatial error model (SEM) (Anselin, 2005). To determine the appropriate model, the Lagrange Multiplier (LM) lag (0.000), LM error (0.000), Robust LM lag (0.000), and Robust LM error (0.052) were tested. The results indicated that SLM was more suitable since it had significant values (<0.05), while Robust LM error was not significant. However, SEM yielded similar outcomes to SLM. An ordinary regression model (OLS) was also employed to assess the performance improvement of SLM, which was measured using the Akaike information criterion (AIC) or Schwarz criterion (SC).

4. Results

The remainder presents the results of the coastal climate risk scores produced by the model and presents a follow up analysis.

4.1. Climate risk score in Jakarta

We applied the model to Jakarta and generated a spectrum of coastal climate risk scores, uncovering the capital's hazards, exposure, and vulnerability at the grid-level. Firstly, the model reports the grid-level map showing the hazard score of riverine and coastal floods (Fig. 3 A, B). The averaged result of hazards from riverine and coastal floods is illustrated in Fig. 3 (C). A score closer to one means that the event has occurred more often and strongly. Fig. 3 (A) depicts that riverine floods have occurred more frequently among riverside areas colored dark blue (check riverside areas in Fig. 3 F).

This result corroborates with monthly flood records constructed by the Jakarta government in 2013 and 2020 (DKI Jakarta, 2013; 2020), which record the extent, height, and length of riverine flood occurred in January 2013 and January 2020. For instance, the *Kampung Melayu* district, which received a high risk score and is situated adjacent to the Ciliwung River—one of the most flood-prone rivers in Jakarta—suffered severe floods in 2013 and 2020. During the 2013 event, 89 % of the district's area was inundated, with an average flood depth of 215 cm over a 13-day period. In 2020, the same area experienced floods with water reaching a depth of 180 cm over three days. In contrast, the Gelora district, located in the central business district, received the lowest score of zero according to our model in 2013 and 2020.

Meanwhile, the extensive northern regions are identified as hazardous due to coastal floods (Fig. 3 B). This is also corroborated by flood records (DKI Jakarta, 2016), which documented the intensity of the coastal flood that occurred in June 2016. The most severely impacted areas were districts in North Jakarta; for instance, the Pluit district experienced flooding with a maximum depth of 50 cm over a period of four days.

It is noteworthy that, in addition to the distinct effects of riverine and coastal floods, there is a possibility of compound floods from the combination of these two types (Bennett et al., 2023). However, the aim of this study is to develop a multi-faceted and lightweight climate risk assessment model based on AI and remote sensing. Therefore, the compound effects are currently calculated by averaging the scores of riverine and coastal floods. Future enhancements could include considering the compound effects through hydrological

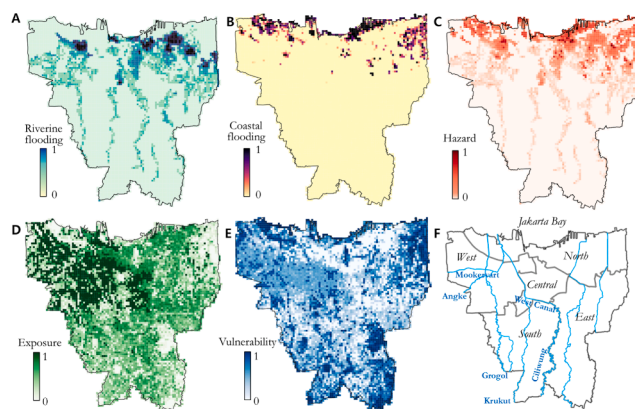


Fig. 3. (A) Riverine flood hazards. (B) Coastal flood hazards. The higher scores indicate the disasters are more possibly to occur among riverside areas and northern coastal areas. The two scores are averaged to produce grid-level hazard score (C). Cluster-based algorithms generate scores for exposure (D) and vulnerability. (F) The administrative divisions and major rivers of Jakarta. While natural disasters occur more in the coastal and riverside areas, the level of exposure is higher in the highly dense residential areas in West Jakarta. Vulnerability scores are higher in the periphery areas near the administrative boundary of Jakarta.

modeling (Ghanbari et al., 2021).

Fig. 3 (D, E) present the grid-level scores of exposure and vulnerability. A high exposure score means that there is a high probability of the presence of people and assets in a grid. The pattern of the exposure map is similar to the population density map (Fig. 2 left, $p = 0.436$) but it is more elaborated as more information from satellite images was put in. Particularly, West and Central Jakarta are shown in dark green with higher exposure scores. The former is area with the highest population density, and the latter is area with central business districts where commercial establishments, hotels, and other crucial assets are clustered. On the other hand, a high vulnerability score means that a grid has less wealth and adaptive capacity to adequately cope with coastal climate hazards. Thus, a grid has a high propensity to be adversely affected by hazards. West Jakarta had the highest score, with an average of 0.504. Meanwhile, Central Jakarta showed a low vulnerability with a white color with an average of 0.430, indicating the central business districts are well prepared with more wealth. The remaining counties (East, South, and North Jakarta) received moderate vulnerability scores, averaging around 0.45 ~ 0.46. The grids with the scores more than 0.9 were observed in the surrounding area of the city center. For example, the grids located on the city boundary in East and South Jakarta have a high agricultural sector and relatively low wealth compared to the urban core.

The three components are aggregated into grid- and district-level risk scores (Fig. 4), where West Jakarta appears as the riskiest in terms of (1) moderate probability of hazard occurrence, (2) high population density that leads to high exposure to hazards, and (3) high level of vulnerability due to relatively insufficient adaptive capacity compared to the urban core. Some districts along the major river, Ciliwung river, showed a high risk score due to the frequent inundation. Conversely, districts in Central Jakarta were given the lowest risk score. Although the districts have a higher exposure score due to their assets, they have been less affected by extreme events and have benefited from their high level of wealth and adaptive capacity to adapt to riverine and coastal floods. This result corroborates with the previous risk assessment studies (Firman et al., 2011:373; Yoo et al., 2014; Yusuf and Francisco, 2009:13), reporting that Central Jakarta is more exposed to the events but are more prepared with extreme events in terms of its high adaptive capacity such as environmental warning systems and humanitarian supporting programs.

Our results corroborate with the qualitative research on high climate risks in West Jakarta (Abidin et al., 2011; Zhang et al., 2022). The high coastal climate risks in West Jakarta is attributed to the compact land use and high slum population ratio. In Jakarta, a slum is measured with eleven variables (BPS Jakarta, 2017). The variables include population density, building density, building materials, condition of ventilation and lighting, toilet condition, waste management, state of drainage, road conditions, public street lighting, and building shape. The density related variables are associated with high exposure in urban slums, and the other variables contributes to high vulnerability of slum residents to climate risk. Particularly, poor water, sanitation, and waste conditions arouse hygienic and health threats during and after floods, which can cause food-, water-, rat-, and mosquito-borne diseases, including skin infection, diarrhea, Leptospirosis, tetanus, and dengue fevers (Yang, 2024). The post-flood diseases became aggravated as the evacuees were exposed to inadequate provision of governmental services (e.g., lighting) and contaminated food and water.

High coastal risks in West Jakarta is also conflated with the poor drainage systems (Kooy and Bakker, 2008; Octavianti and Charles, 2019). The city center with a high proportion of commercial areas has enjoyed the drains constructed since the colonial area. For example, the colonial government constructed a canal named Mookervart to provide a water source outside the city center by the late 17th century. Another significant drain constructed during the colonial era was *bandjir kanaal* (West Flood Canal). After suffering from major floods in the 20th century, the government constructed *bandjir kanaal* to carry river water from the city center to other surrounding areas including West Jakarta. The canal was successful in preventing floodwater from the old city center. However, West Jakarta became a bottleneck of significant rivers (Angke, Grogol, Krukut, and Ciliwung). As a result, West Jakarta and residential areas with slum communities are constantly threatened by flooding. Several pump stations were installed to drain out the stagnant water in those areas; however, the pumps are not enough to handle a large number of slum and vulnerable communities.

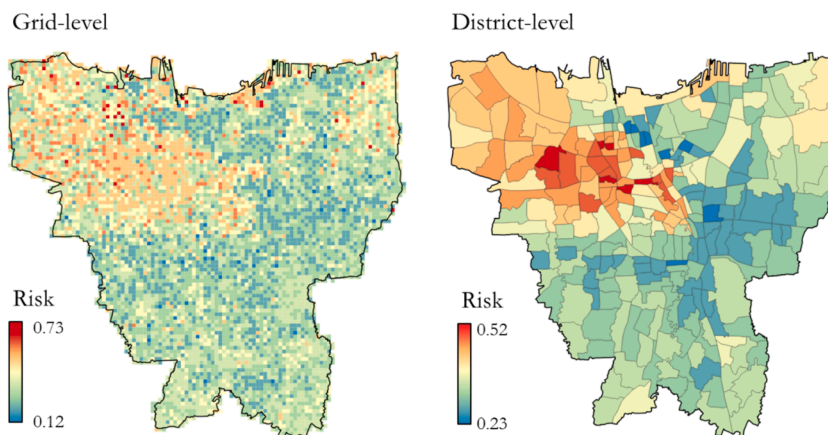


Fig. 4. Coastal climate risk scores at grid (left) and district (right) levels.

4.2. Performance evaluation

Table 2 reports the correlation values between the predicted scores from four models and ground-truth labels. The evaluation based on GIS footprints including buildings and roads are performed both at grid- and district-levels. Whilst, those with municipal statistics are evaluated solely at district-level. The evaluation calculates spearman correlations on the log-scaled ground-truth values because our model is targeted to produce ranking-based scores. For exposure scores, the evaluation compares our model to GT regression and pairwise models trained with population data, and for vulnerability scores, it compares models trained with RWI.

Our model produced the best performance among six out of eight validation items. Also, it generated scores of solid correlations with all items (i.e., above 0.4). Only on “the grid-level road length” item did the pairwise model perform better than our model at both grid- and district-levels. However, the difference in performance was not significant (e.g., grid-level, pairwise: 0.8551; our model: 0.8503). It is meaningful to note that our model outperformed the model using ground-truth directly (i.e., GT regression models). This outperformance suggests that the model increased the predictive power by learning additional information shown in the satellite image. In addition, our cluster-based model consistently outperformed the pairwise model, particularly for district-level items, by effectively leveraging the abundant hierarchical relationship information in the ranking of clusters (e.g., pairwise: 0.0258; our model: 0.6508).

4.3. Spatial regression model

We next examine which variables are associated with coastal climate risks with spatial regression model. The result from OLS and SLM is presented in Table 3. SLM reported 0.738 for the coefficient of determination (R^2), AIC for -971.2 , and SC for -903.4 . These figures show better performance of SLM by adding a spatial lag term to the OLS model whose R^2 is 0.468, AIC is -827.4 , and SC is -763.2 . As R^2 for spatial regression models is pseudo- R^2 , it is more appropriate to confirm AIC or SC as measures of the model strength. The lower AIC and BIC indicated a better model fit. Moreover, the p-value for the spatial lag term (ρ) is less than 0.005. The p-value reconfirms that there is significant spatial autocorrelation for the dependent variable and that the OLS regression model would be misleading.

Several variables appear significant in SLM, including the slum rate ($p < 0.005$). Higher slum conditions indicate higher risks in terms of coastal climate. The coefficient of religion area is negatively significant ($p = 0.044$). This means the areas with more religion buildings are well prepared and less susceptible to climate risk. Districts with more drainage systems is relatively less risky ($p = 0.009$) in OLS. However, all flooding mitigation facilities have no significant results in SLM. There have been several governmental efforts in the 2010 s to mitigate riverine flood hazards, including the Ciliwung River normalization projects, the construction of the Manggarai flood gate, and the addition of dams in the upstream area of Jakarta. Additionally, most of Jakarta’s estuaries are equipped with flood gates to mitigate coastal floods. Nevertheless, although these measures are expected to impact flood control (e.g., Yatsrib et al., 2021), they do not show a significant relationship with the overall climate risk score, which includes factors such as population exposure and local wealth levels. This result suggests that efforts to address urban poverty and excessive population density need to be considered a primary governmental aim, alongside the construction of infrastructure.

5. Discussion and conclusion

We presented a machine learning-based computation of climate risk and its application on Jakarta. Our model allows for a multi-dimensional assessment by quantifying hazards, exposure, and vulnerability (IPCC, 2022). The model measured not only the extent of natural disasters from satellite imagery but also the sensitivity and adaptive capacities of a region to climate change. The evaluation revealed that the cluster-based approach produces more precise outcomes to predict risks than using grid-level labels directly (Table 2). The spatial regression model on district-level scores examined that the most marginalized communities, such as slum settlements, are at the greatest risks of the impacts of riverine and coastal floods in Jakarta (Table 3).

Our model has practical implications for future humanitarian projects and international efforts to address climate risks. First, the lightweight model design can run efficiently on a variety of computing platforms, reducing the time and resources required to generate results, making it attractive for non-profit organizations and developing countries with limited budget (Ahn et al., 2023). In addition,

Table 2

Comparison of scoring models for climate risks in Jakarta. Performance is evaluated both at grid- and district levels with building density, road length, population density, the number of evacuees during the 2013 flood, the number of employments, and the amount of government expenditure. The table reports the spearman correlation values between the estimated scores and ground-truth labels.

Level	Component	Proxy	GT regression	NL Regression	Pairwise	Our model
Grid-level	Exposure	Building density	0.6840	0.6773	0.7052	0.7142
		Road length per area	0.8160	0.7932	0.8551	0.8503
District-level	Exposure	Population density	0.5753	0.1209	0.3618	0.7221
		Building density	0.7421	0.1963	0.5064	0.7653
		Road length per area	0.4032	0.5262	0.4335	0.4835
		Number of evacuees(2013 flood)	0.3146	0.0956	0.2579	0.4330
		Employment per area	0.5755	0.1298	0.0258	0.6508
	Vulnerability	Expenditure per area	0.5572	0.5504	0.1974	0.5769

Table 3

Ordinary least squares (OLS) and Spatial lag model (SLM) of coastal climate risk scores on socio-economic variables (N=262). Slum ratio, commercial area, and selected types of flood facilities have significant relationships with the scores.

Variables	OLS		SLM	
	Coefficient	p-value	Coefficient	p-value
Spatial lag (ρ)			0.785	0.000***
Constant	0.354	0.000***	0.089	0.001***
Slum rate (%)	0.105	0.005**	0.071	0.004***
Unemployment (%)	-0.207	0.305	-0.108	0.427
Commercial area (%)	-0.099	0.038*	-0.043	0.181
Industrial area (%)	-0.018	0.666	0.014	0.617
Public area (%)	-0.059	0.427	-0.094	0.062
Religion area (%)	-1.267	0.038*	-0.826	0.044*
Medical area (%)	0.321	0.120	0.174	0.212
Education area (%)	0.300	0.117	0.245	0.058
Green area (%)	-0.013	0.758	-0.010	0.740
Floodgate (dummy)	-0.010	0.405	-0.005	0.496
Flood post (# / km ²)	0.010	0.080	0.004	0.341
Pump station (# / km ²)	-0.002	0.458	-0.004	0.075
Drainage (km / km ²)	-0.001	0.009**	-0.000	0.514
County fixed effects	Yes		Yes	
R-squared	0.468		0.738	
AIC & SC	-827.4 & -763.2		-971.2 & -903.4	
DF	244		243	

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.005$.

the low-cost approach to data acquisition and processing, utilizing readily available satellite imagery, allows for a wider access to climate assessment in regions where data collection and analysis have previously been prohibitively expensive.

Another key strength of the model is its ability to provide grid-level assessment of climate risk. While existing national-scale reports provide useful information, they may not capture the full range of local conditions and factors (Adger et al., 2018). In contrast, the model presented here utilizes one-meter resolution satellite imagery to generate detailed information about socio-economic conditions. This granular-level approach provides valuable insights into the unique risk factors affecting individual communities, allowing for more targeted decision-making (see [Supplementary Materials](#) for the comparison of our model with other previous assessment). For example, a slum area with a low hazard score and a high vulnerability score may require a different policy approach than a coastal area with a high hazard score.

There are numerous ways our work can be extended in the future. One is to better define the risk components. For instance, our hazard score only considers natural events, yet it can also expand to include human-induced events such as anthropogenic land subsidence. We can also consider more accurately defining exposure and vulnerability, for instance, by including the education level of individuals and public campaigns for increasing awareness. Another future direction is to tune the ratio of risk components among hazards, exposure, and vulnerability. We have considered an equal weight in combining these scores, yet the impact of interest may differ for different regions and analysis tasks. Future research can look into various weight assignments.

Moreover, our work is currently limited to measuring other climate hazards. Subsequent research can extend the model's application to include additional climate risks, such as droughts and heatwaves, to provide a more comprehensive framework. Lastly, satellite-based assessment cannot fully forecast upcoming climate risks because images cannot capture future events. Our study focuses on existing vulnerabilities and exposures, providing an estimate of the current predisposition to natural disasters. Future climate risks can be more comprehensively assessed by integrating our results with projections on socio-economic conditions and climate change scenarios, such as predictions of rainfall trends.

6. Declaration of Generative AI in Scientific Writing

During the preparation of this work the authors used ChatGPT in order to proofread the draft. After using ChatGPT, the authors reviewed and edited the content as needed and take full responsibility for the content of the publication.

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CRediT authorship contribution statement

Jeasurk Yang: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Donghyun Ahn:** Writing – review & editing, Project administration, Conceptualization. **Junbeom Bahk:** Writing – review & editing, Methodology, Investigation, Conceptualization. **Sungwon Park:** Writing – review & editing, Validation, Software, Methodology, Formal analysis,

Conceptualization. **Nurrokhmah Rizqihandari**: Writing – review & editing, Project administration, Conceptualization. **Meeyoung Cha**: Writing – review & editing, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

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References

- Abidin, H.Z., Djaja, R., Darmawan, D., Hadi, S., Akbar, A., Rajiyowiryo, H., Sudibyo, Y., Meilano, I., Kasuma, M.A., Kahar, J., Subarya, C., 2001. Land subsidence of Jakarta (Indonesia) and its geodetic monitoring system. *Natural Hazards* 23, 365–387.
- Abidin, H.Z., Andreas, H., Gumilar, I., Fukuda, Y., Pohan, Y.E., Deguchi, T., 2011. Land subsidence of Jakarta (Indonesia) and its relation with urban development. *Natural Hazards* 59, 1753–1771.
- Abrams, M., Bailey, B., Tsu, H., Hato, M., 2010. The ASTER Global DEM. *Photogrammetric Engineering and Remote Sensing* 76 (4), 344–348.
- Adger, W.N., Brown, I., Surminski, S., 2018. Advances in risk assessment for climate change adaptation policy. *Philosophical Transactions of the Royal Society a: Mathematical, Physical and Engineering Sciences* 376 (2121), 20180106.
- Ahn, D., Yang, J., Cha, M., Yang, H., Kim, J., Park, S., Han, S., Lee, E., Lee, S., Park, S., 2023. A human-machine collaborative approach measures economic development using satellite imagery. *Nature Communications* 14, 6811.
- Anselin, L., 2005. *Exploring Spatial Data with Geoda: A Workbook*. University of Illinois Urbana, Urbana Champaign.
- Antzoulatos, G., Koutoglou, I.O., Bakratsas, M., Moutzidou, A., Gialampoukidis, I., Karakostas, A., Lombardo, F., Fiorin, R., Norbiato, D., Ferri, M., Symeonidis, A., Vrochidis, S., Kompatsiaris, I., 2022. Flood hazard and risk mapping by applying an explainable machine learning framework using satellite imagery and GIS data. *Sustainability* 14 (6), 3251.
- Arribas, A., Farigrieve, R., Dhu, T., Bell, J., Cornforth, R., Gooley, G., Hilson, C.J., Luers, A., Shepard, T.G., Street, R., Wood, N., 2022. Climate risk assessment needs urgent improvement. *Nature Communications* 13 (1), 1–4.
- Bennett, W.G., Karunaratna, H., Xuan, Y., Kusuma, M.S.B., Farid, M., Kuntoro, A.A., Rahayu, H.P., Kombaitan, B., Septiadi, D., Kesuma, T.N.A., Haigh, R., Amaratunga, D., 2023. Modelling compound flooding: a case study from Jakarta, Indonesia. *Natural Hazards* 118, 277–305.
- BIG (2024) Badan Informasi Geospasial – Official website. Available at: <https://tanahair.indonesia.go.id/demnas/#/> (Accessed August 10th 2024).
- BPS Jakarta Barat, 2021. Kota Administrasi Jakarta Barat Dalam Angka 2021: Jakarta Barat Municipality in Figures. Badan Pusat Statistik Jakarta Barat Municipality, Jakarta.
- BPS Jakarta Pusat, 2021. Kota Administrasi Jakarta Pusat Dalam Angka 2021: Jakarta Pusat Municipality in Figures. Badan Pusat Statistik Jakarta Pusat Municipality, Jakarta.
- BPS Jakarta Selatan, 2021. Kota Administrasi Jakarta Selatan Dalam Angka 2021: Jakarta Selatan Municipality in Figures. Badan Pusat Statistik Jakarta Selatan Municipality, Jakarta.
- BPS Jakarta Timur, 2022. Kota Administrasi Jakarta Timur Dalam Angka 2022: Jakarta Timur Municipality in Figures. Badan Pusat Statistik Jakarta Timur Municipality, Jakarta.
- BPS Jakarta Utara, 2021. Kota Administrasi Jakarta Utara Dalam Angka 2021: Jakarta Utara Municipality in Figures. Badan Pusat Statistik Jakarta Utara Municipality, Jakarta.
- Budyono, Y., Aerts, J., Brinkman, J., Marfai, M.A., Ward, P., 2015. Flood risk assessment for delta mega-cities: a case study of Jakarta. *Natural Hazards* 75, 389–413.
- Cao Z, Qin T, Liu TY, Tsai MF and Li H (2007) Learning to rank: from pairwise approach to listwise approach. In: *Proceedings of the 24th international conference on Machine learning* (ed Z Ghahramani), Corvallis Oregon, USA, 20–24 June 2007, pp. 129–136. New York: Association for Computing Machinery.
- Caron M, Bojanowski P, Joulin A and Douze M (2018) Deep clustering for unsupervised learning of visual features. In: *Proceedings of European Conference on Computer Vision (ECCV)* (eds V Ferrari, M Herbert, C Sminchisescu and Y Weiss), Munich, Germany, 8–14 September 2018, pp. 132–149. Cham: Springer.
- Chaussard, E., Amelung, F., Abidin, H., Hong, S.H., 2013. Sinking cities in Indonesia: ALOS PALSAR detects rapid subsidence due to groundwater and gas extraction. *Remote Sensing of Environment* 128, 150–161.
- Deltares (2024) *1D/2D modelling suite for integral water solutions SOBEK*. Deltares, Delft, available online at: https://content.oss.deltares.nl/sobek2/SOBEK_User_Manual.pdf (Accessed August 10th 2024).
- Elvidge, C.D., Zhizhin, M., Ghosh, T., Hsu, F.-C., Taneja, J., 2021. Annual time series of global VIIRS nighttime lights derived from monthly averages: 2012 to 2019. *Remote Sensing* 13 (5), 922.
- Engilberge M, Chevallier L, Perez P and Cord M (2019) SoDeep: A sorting deep net to learn ranking loss surrogates. In: *2019 IEEE/CVF Conference on Computer Vision and Pattern Recognition* (eds L Davis, P Torr and S-C Zhu), Long Beach, USA, 15–20 June 2019, pp. 10784–10793. Los Alamitos: IEEE Computer Society.
- Firman, T., Surbakti, I.M., Idroes, I.C., Simarmata, H.A., 2011. Potential climate-change related vulnerabilities in Jakarta: Challenges and current status. *Habitat International* 35, 372–378.
- Ghanbari, M., Arabi, M., Kao, S., Obeyseker, J., Sweet, W., 2021. Climate change and changes in compound coastal-riverine flooding hazard along the U.S. coasts. *Earth's Future* 9 e2021EF002055.

- Gupta R, Goodman B, Patel N, Hosfelt R, Sajeev S, Heim E, Doshi J, Lucas K, Choset H and Gaston M (2019) Creating xBD: A dataset for assessing building damage from satellite imagery. In: *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition Workshops 2019* (ed L O'Conner), Long Beach, USA, 16–20 June 2019, pp. 10–17. Los Alamitos: IEEE Computer Society.
- Hakam, O., Baali, A., Belhaj Ali, A., 2023. Modeling drought-related yield losses using new geospatial technologies and machine learning approaches: case of the Gharb plain, North-West Morocco. *Modeling Earth Systems and Environment* 9 (1), 647–667.
- Han S, Ahn D, Cha H, Yang J, Park S and Cha M (2020a) Lightweight and robust representation of economic scales from satellite imagery. *Proceedings of the AAAI Conference on Artificial Intelligence* 34(1): 428–436.
- Han S, Ahn D, Park S, Yang J, Lee S, Kim J, Yang H, Park S and Cha M (2020b) Learning to score economic development from satellite imagery. In: *Proceedings of the 26th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining* (eds J Tang and BA Prakash), Virtual Event, 6–10 July 2020, pp. 2970–2979. New York: Association for Computing Machinery.
- Henderson, J.V., Storeygard, A., Weil, D.N., 2012. Measuring economic growth from outer space. *American Economic Review* 102, 994–1028.
- IPCC (2022) *Climate Change 2022: Impacts, Adaptation, and Vulnerability. Contribution of Working Group II to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*. Pörtner HO, Roberts DC, Tignor M, Poloczanska ES, Mintenbeck K, Alegría A, Craig M, Langsdorf S, Lösschke S, Möller V, Okem A and Rama B (eds.) Cambridge and New York: Cambridge University Press.
- Jakarta, D.K.I., 2013. Data Kejadian Bencana Banjir Tahun 2013 di DKI Jakarta. BPBD Provinsi DKI Jakarta, Jakarta.
- Jakarta, D.K.I., 2016. Data Kejadian Bencana Banjir Tahun 2016 di DKI Jakarta. BPBD Provinsi DKI Jakarta, Jakarta.
- Jakarta, B.P.S., 2017. Pendataan RW Kumuh DKI Jakarta 2017. Badan Pusat Statistik Provinsi DKI Jakarta, Jakarta.
- Jakarta, D.K.I., 2020. Data Kejadian Bencana Banjir Tahun 2016 di DKI Jakarta. BPBD Provinsi DKI Jakarta, Jakarta.
- Jean, N., Burke, M., Xie, M., Davis, W.M., Lobell, D.B., Ermon, S., 2016. Combining satellite imagery and machine learning to predict poverty. *Science* 353 (6301), 790–794.
- Jones, R.N., 2001. An environmental risk assessment/management framework for climate change impact assessments. *Natural Hazards* 23 (2–3), 197–230.
- Kim, D., Won, J., Lee, E., Park, K., Kim, J., Park, S., Yang, H., Cha, M., 2022. Disaster assessment using computer vision and satellite imagery: Applications in water-related building damage detection. *Frontiers in Environmental Science* 10, 969758.
- Kooy, M., Bakker, K., 2008. Splintered networks: The colonial and contemporary waters of Jakarta. *Geoforum* 39, 1843–1858.
- Leichenko, R., Silva, J.A., 2014. Climate change and poverty: vulnerability, impacts, and alleviation strategies. *Wires Climate Change* 5, 539–556.
- López Royo, M., Ranasinghe, R., Jiménez, J.A., 2016. A rapid, low-cost approach to coastal vulnerability assessment at a national level. *Journal of Coastal Research* 32 (4), 932–945.
- Verisk Maplecroft (2021) *Environmental Risk Outlook 2021*. Available at: <https://www.maplecroft.com/insights/analysis/environmental-risk-outlook-2021/> (Accessed August 10th 2024).
- Mateo-García, G., Veitch-Michaelis, J., Smith, L., Oprea, S.V., Schumann, G., Gal, Y., Baydin, A.G., Backes, D., 2021. Towards global flood mapping onboard low cost satellites with machine learning. *Scientific Reports* 11 (1), 1–12.
- Menard, S., 2001. *Applied Logistic Regression Analysis*, 2nd Edition. Sage Publications, Thousand Oaks.
- Meta (2024) Relative Wealth Index. Available at: <https://dataforgood.facebook.com/dfg/tools/relative-wealth-index> (Accessed August 10th 2024).
- Mohan, A., Singh, A.K., Kumar, B., Dwivedi, R., 2021. Review on remote sensing methods for landslide detection using machine and deep learning. *Transactions on Emerging Telecommunications Technologies* 32 (7), e3998.
- Mokrech, M., Nicholls, R.J., Dawson, R.J., 2012. Scenarios of Future Built Environment for Coastal Risk Assessment of Climate Change Using a GIS-Based Multicriteria Analysis. *Environment and Planning b: Planning and Design* 39 (1), 120–136.
- Octavianti, T., Charles, K., 2019. The evolution of Jakarta's flood policy over the past 400 years: The lock-in infrastructural solutions. *Environment and Planning c: Politics and Space* 37 (6), 1102–1125.
- Open data Jakarta, 2024. Open data Jakarta – Official website. Available at: <https://data.jakarta.go.id/> (Accessed August 10th 2024).
- OpenStreetMap, 2024. Open Street Map Indonesia – Official website. Available at: <https://openstreetmap.or.id/> (Accessed August 10th 2024).
- Pantau Banjir Jakarta (2024) Pantau Banjir Jakarta – Official website. Available at: <https://pantaubanjir.jakarta.go.id/> (Accessed August 10th 2024).
- Peng, J., Zhang, J., 2022. Urban flooding risk assessment based on GIS-game theory combination weight: A case study of Zhengzhou City. *International Journal of Disaster Risk Reduction* 77, 103080.
- Rolnick, D., Donti, P.L., Kaack, L.H., Kochanski, K., Lacoste, A., Sankaran, K., Ross, A.S., Milojevic-Dupont, N., Jaques, N., Waldman-Brown, A., Luccioni, A.S., Maharaj, T., Sherwin, E.D., Mukkavilli, S.K., Kording, K.P., Gomes, C.P., Ng, A.Y., Hassabis, D., Platt, J.C., Creutzig, F., Chayes, J., Bengio, Y., 2022. Tackling climate change with machine learning. *ACM Computing Surveys (CSUR)* 55 (2), 1–96.
- Jakarta Satu (2024) Jakarta Satu – Official website. Available at: <https://jakartasatu.jakarta.go.id/> (Accessed August 10th 2024).
- Scott, P., 2016. How climate change affects extreme weather events. *Science* 352 (6293), 1517–1518.
- Simarmata, H.A., Surtiari, G.A.K., 2019. *Adaptation to Climate Change: Decision Making and Opportunities for Transformation in Jakarta, Indonesia*. UNRISD, Geneva.
- Takagi, H., Esteban, M., Mikami, T., Fujii, D., 2016. Projection of coastal floods in 2050 Jakarta. *Urban Climate* 17, 135–145.
- UNDRR (2019) *Global Assessment Report on Disaster Risk Reduction 2019*. Geneva: United Nations Office for Disaster Risk Reduction (UNDRR).
- Wijayanti, P., Zhu, X., Hellegers, P., Budiyo, Y., van Ierland, E.C., 2017. Estimation of river flood damages in Jakarta, Indonesia. *Natural Hazards* 86, 1059–1079.
- WorldPop (2024) WorldPop – Official website. Available at: <https://www.worldpop.org/> (Accessed August 10th 2024).
- Wu, C., Zhang, F., Xia, J., Xu, Y., Li, G., Xie, J., Du, Z., Liu, R., 2021. Building damage detection using u-net with attention mechanism from pre-and post-disaster remote sensing datasets. *Remote Sensing* 13 (5), 905.
- Xie B, Xu J, Jung J, Yun SH, Zeng E, Brooks EM, Dolk M and Narasimhalu L (2020) Machine Learning on Satellite Radar Images to Estimate Damages After Natural Disasters. In: *Proceedings of the 28th International Conference on Advances in Geographic Information Systems* (eds CT L, F Wang, G Trajcevski, Y Huang, S Newsam and L Xiong), Seattle, USA, 3–6 November 2020, pp. 461–464. New York: Association for Computing Machinery.
- Yang, J., 2024. Waste accumulation in Jakarta's slums: Neoliberal flows of waste distribution. *Geoforum* 150, 103994.
- Yatsrib, M., Harman, A.N., Taufik, S.R., Kesuma, T.N.A., Saputra, D., Kusuma, M.S.B., Farid, M., Kuntoro, A.A., 2021. Study on the contribution of normalization to reducing flood risk in the Ciliwung river, Tebet district, Jakarta. *IOP Conference Series: Earth and Environmental Science* 933 (1), 012032.
- Yeh, C., Perez, A., Driscoll, A., Azzari, G., Tang, Z., Lobell, D., Ermon, S., Burke, M., 2020. Using publicly available satellite imagery and deep learning to understand economic well-being in Africa. *Nature Communications* 11 (1), 1–11.
- Yoo, G., Kim, A.R., Hadi, S., 2014. A methodology to assess environmental vulnerability in a coastal city: Application to Jakarta, Indonesia. *Ocean & Coastal Management* 102, 169–177.
- Yusuf AA and Francisco H (2009) *Climate Change Vulnerability Mapping for Southeast Asia*. Singapore: Economy and Environment Program for Southeast Asia (EEPSEA).
- Zennaro, F., Furlan, E., Simeoni, C., Torresan, S., Aslan, S., Critto, A., Marcomini, A., 2021. Exploring machine learning potential for climate change risk assessment. *Earth-Science Reviews* 220, 103752.
- Zhang, H., Liu, X., Xie, Y., Gou, Q., Li, R., Qiu, Y., Hu, Y., Huang, B., 2022. Assessment and improvement of urban resilience to flooding at a subdistrict level using multi-source geospatial data: Jakarta as a case study. *Remote Sensing* 14 (9), 2010.
- Zscheischler, J., Westra, S., van den Hurk, B.J.J.M., Seneviratne, S.I., Ward, P.J., Pitman, A., Aghakouchak, A., Bresch, D.N., Leonard, M., Wahl, T., Zhang, X., 2018. Future climate risk from compound events. *Nature Climate Change* 8, 469–477.